

CLASS 9th
MATHS ADVANCE

A. Algebra

1. If $x + y = 5$ and $xy = 6$, find $x^3 + y^3$
2. Solve: $x^4 - 5x^2 + 4 = 0$

B. Geometry

1. Prove that in a right triangle, the median on hypotenuse = half of hypotenuse
2. A quadrilateral ABCD is a parallelogram. Diagonal AC = 10 cm, angle at A = 60° . Find length of diagonal BD.

C. Coordinate Geometry

1. Find equation of line passing through (2, 3) and perpendicular to line $3x - 4y + 7 = 0$
2. Find area of triangle formed by points (1, 2), (4, 6), (5, 1)

D. Mensuration

1. A metallic sphere of radius 7 cm is melted to form smaller spheres of radius 1 cm. Find number of smaller spheres.
2. A frustum of cone has radii 7 cm and 14 cm, height 9 cm. Find volume.

E. Statistics & Probability

1. Find variance and standard deviation of: 2, 4, 4, 4, 5, 5, 7, 9
2. A bag has 3 red, 4 green, 5 blue balls. Find probability of drawing:
 - (a) Red or Blue
 - (b) Not Green